

In the world of AI, it seems that 2025 is the year of the AI Agent.

New agentic workflows and models are released all the time,

often accompanied by breathless declarations on social media that a task that previously required human expertise

has now been entirely automated by the latest agentic breakthrough.

But can you distinguish a simple reflex agent from an advanced learning agent?

You see agents are classified based on their level of intelligence,

based on their decision-making processes and how they interact with their surroundings to reach wanted outcomes.

So let's explore the five main types of AI agents to understand what they can and cannot do.

Simple Reflex Agent

Now a simple reflex agent that is the most simple type of AI agent,

the most basic type, and it follows predefined rules to make decisions like a thermostat.

It turns on the heat when the temperature drops below a predefined threshold and then it turns it off again when a set temperature is reached.

So let's break it down.

We've got our agent here.

Now the the environment over here that's the external world that the agent is embedded into and it next to.

Then we've got precepts.

These are the perceived input from the environment as measured through sensors.

Then these sensors feed the precept into the internal logic of the agent

which gives us a representation of what the world is like now,

and knowing what the world is like now we can use condition action rules as the core logic of these simple reflex agent.

Now, these are rules that follow an if condition.

Then action structure.

So if the temperature drops to 18 Celsius then turn on the heat.

That's executed by actuators and that results in an action.

The output behavior by the agent and that action affects the environment which in turn affects the next set of precepts and around and around we go.

Simple reflex agents like this are effective in structured

and predictable environments where the rules are well defined, but dynamic scenarios?

They can trip these agents up,

and because they don't store past information, they can repeatedly make the same mistakes

if the predefined rules are insufficient for handling new situations.

Model-Based Reflex Agent

All right, well how about this one?

This is called a model based reflex agent.

So this is a more advanced version of the the simple reflex agent,

and it uses condition action rules to make decisions as well but,

it also incorporates an internal model of the world and that's stored in the state component and that state component is updated by observing how the world actually evolves.

Essentially how the environment changes from one state to another.

The agent also tracks how its own actions affect the environment.

That's what my actions do.

And all of this is used instead of just taking the raw precepts data for decision making.

So take a robotic vacuum cleaner for example.

The internal state that remembers where it's been and what areas are clean and where the obstacles are.

It knows that if it moves forwards, it changes its location and that action has consequences.

And it has condition-action rules, like if I think I'm in a dirty area and I haven't cleaned it yet, then vacuum it.

It doesn't just react to what it immediately sees, it infers and it remembers parts of the environment it can't currently observe.

That's model-based reasoning in action.

Goal-Based AI Agent

now a goal-based AI model that is building on top of the model-based agent by adding decision-making that's based on goals.

So we don't have any more condition action rules, we have goals, and they represent the desired output the agent is trying to achieve.

So the agent uses its model, that's how the world evolves and what my actions do, to simulate future outcomes of possible actions, essentially predicting what will it be like if I do action A.

Now that's a shift in decision making.

The agent isn't just asking what action matches this condition, it's now asking what actually will help me achieve my goal based on the current state and predicted future.

So consider a self-driving car.

If the goal is to get to destination X, It'll consider its state, which is, I'm on Main Street.

It will then generate a prediction.

If I turn left, I'll head towards the highway, and it'll ask, will that help me reach destination X?

And if the answer is yes, then the action will be to turn left.

Goal-based agents are widely used in robotics and simulations where a clear objective is set and adaptation to the environment is required.

Utility Based AI Agent

Now a utility-based agent looks like this.

And it considers not just if a goal is met, but how desirable different outcomes are.

So utility here represents a happiness score or a preference value for a particular outcome.

So for each possible future state, the agent asks how happy will I be in such a state or really the expected utility of the future state.

And this lets it rank options, not just pick anything that meets the goal.

So consider an autonomous drone delivery.
The goal-based version might be to use a goal of deliver the package to address X, and it chooses an action that completes that goal. Doesn't matter if it gives you a bumpy energy-wasting route, but a utility-based person, that might instead be something like deliver the packages quickly and safely and with minimum energy usage, whereby now the drone simulates multiple paths, it estimates things like duration and battery level and weather, and it picks the route that maximizes its utility score. That's AI agent number four.

Learning AI Agent

Now, the fifth agent is the most adaptable and also the most powerful and it is the learning agent. So rather than being hard coded or being goal driven, it learns from experience. It improves its performance over time by updating its behavior based on feedback from the environment. So how does it work? There's a critic component and that observes the outcome of an agent's actions via the sensors and it compares them to a performance standard. Now that gives us a numerical feedback signal that's often called a reward in reinforcement learning and this reward is then passed to a learning element that updates the agent's knowledge using the feedback from the critic. Its job here is to improve the agents mapping from states all the way through to actions. Now the problem generator, that suggests new actions the agent hasn't tried yet, like try a different path, see if it's any faster. And then the performance element selects actions based on what the learning element has determined to be optimal. So think of an AI chess bot, the performance elements that plays the game using current learn strategies. The critic, you'll see that it lost the match. The learning element adjusts its strategy based on the outcomes of thousands of games and the problem generator suggests new moves that it hasn't explored yet.

Use Cases

So a simple reflex agent reacts. It's fast to execute but it has no memory and it has no understanding of history. A model-based reflex agent, we can really think that the difference there is that that remembers. It does that by tracking state over time. It doesn't plan, it's still reactive. Now a goal-based model, that aims. It aims by using goal-directed behavior but any way of meeting that goal...

Will do.

Whereas, an utility-based agent that takes a different path, it evaluates.

It does that by choosing the best outcome, but requires an accurate utility function to do so.

And then a learning agent that improves by learning from experience,

but this can be the slowest and most data intensive process.

Now in many cases, we will want to use multiple agents together.

That is called a multi-agent system.

And that's where multiple agents operate in a shared environment,

working in a cooperative way, working towards a common goal.

And as agentic AI continues to evolve, particularly with learning agents that are making uses of advances in generative AI,

AI agents are becoming increasingly adept at handling complex use cases,

but it's not really all over for us just yet.

AI agents typically work best with a good old human in the loop.

At least for the time being.